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B.TECH, C.E., MAJOR PROJECT - PHASE - I

AN APPROACH TOWARDS BUILDING A SMART LIFT SYSTEM USING MACHINE LEARNING

PROJECT GUIDE
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AT PRESENT...

- Lift systems heavily rely on the pre-programmed behaviour
- Lack of real-time adaptability to user patterns
- Increased energy consumption due to static operation
- Longer wait times for users during peak demand periods
- Buildings with multiple lifts only worsen the energy situation
- Difficulty in optimizing routes and energy usage

PROBLEM DEFINITION

In modern buildings, lift systems are essential for efficient transportation across floors. However, conventional lift systems often operate on basic pre-programmed logic, leading to inefficiencies such as prolonged wait times, unnecessary stops, and high energy consumption. This project aims to develop a smart lift system using machine learning techniques to optimize lift operations. By predicting future lift calls based on historical usage patterns and current conditions, the system will improve the overall efficiency, reducing wait times and energy consumption. The model will be trained on synthesized time-series data to anticipate user demands and optimize lift movement.

- How to make the lift system ‘know’ where the next request is going to be from? - Using RNNs with Time-Series Data

SCOPE

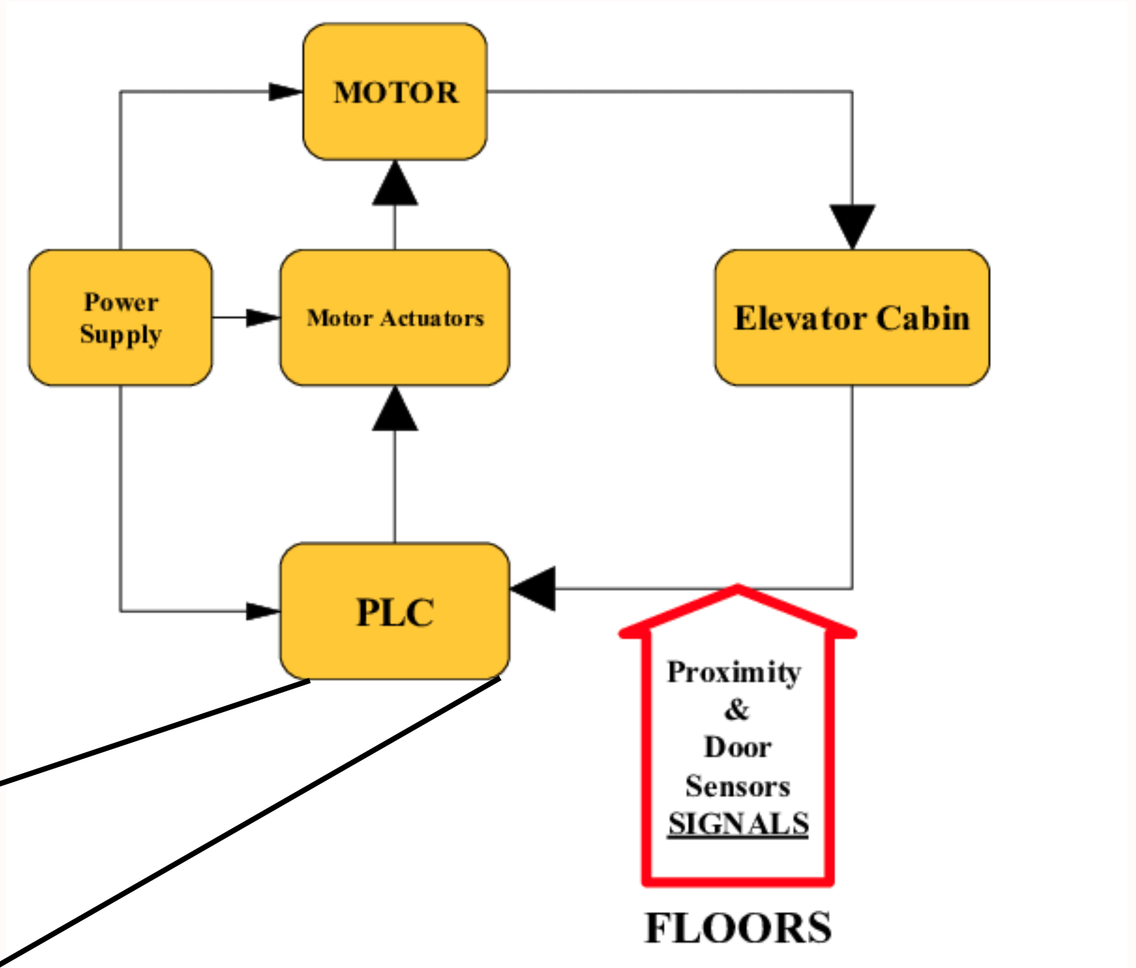
- Under consideration - Residential lifts, Office Building lifts.
- Analyze and understand historical lift usage patterns to identify inefficiencies.
- Generate synthetic time-series data to simulate lift usage for model training.
- Build and train predictive models to forecast future lift calls and optimize movement.
- Design and develop a dedicated simulation software to simulate various lift models for boosting the research.
- Measure improvements in terms of reduced wait times, optimized routing, and energy savings.
- Ensure scalability for various building sizes and adaptability to different usage patterns.
- Improve user experience by minimizing delays and unnecessary stops.

PROJECT OBJECTIVES

- To enhance user experience by minimizing wait times for the lift.
- To design lift systems that optimize power consumption and reduce overall energy usage.
- To predict lift usage patterns using machine learning for efficient routing and operations.
- To develop a scalable system adaptable to different building sizes and usage scenarios.
- To reduce unnecessary lift stops and improve the overall efficiency of lift movement.
- To integrate data-driven decision-making for smarter, real-time lift operations.

LITERATURE REVIEW

- Current lift systems manage lift calls through PLC (Programmable Logic Controller)
- PLC is coded and followed as a protocol
- Introducing a ‘smart’ component over the PLC may help optimize power usage and reduce wait times



Current Lift System Schematic

AT A GLANCE...

- Hochreiter and Schmidhuber (2020) introduced Long Short-Term Memory (LSTM) networks, which have been shown to effectively model time-dependent sequences, making them highly suitable for predicting elevator usage patterns based on historical data.
- Saha and Senapati (2020) demonstrated the power of LSTM networks in analyzing sequential data, which can be applied to optimize lift system operations by anticipating future demands and adjusting in real time.
- Sherstinsky (2020) highlighted the fundamentals of Recurrent Neural Networks (RNNs) and LSTM, providing a robust framework for capturing complex time-series dependencies in smart elevator systems.
- Rezk et al. (2020) provided insights into the use of RNNs in embedded systems, showing that lightweight, real-time implementations of machine learning models can be achieved on resource-constrained platforms like smart elevators.
- Al-Kodmany (2023) discussed the role of smart elevator systems in modern buildings, emphasizing the integration of AI technologies to enhance system efficiency and reduce energy consumption.

POWER CALCULATIONS

2 ESTIMATED DAILY ENERGY CONSUMPTION

The estimated daily energy consumption (E_d) of a lift is the sum of the running consumption (E_{rd}) and the standing (idle/standby) consumption (E_{sd}):

$$E_d = E_{rd} + E_{sd} \quad \ldots (1)$$

$$E_{rd} = \frac{n_d \times \%S \times k_L \times E_{rc}}{2} \quad \ldots (2)$$

where:

n_d is the number of trips per day[4]
$\%S$ is the percentage average travel distance per trip for a target installation
k_L is the load factor per trip
E_{rc} is the measured or estimated running energy consumption of the ISO reference cycle (two trips) in Wh.

$$E_{sd} = (24 - \frac{n_d}{3600} t_{av})(P_{id} R_{id} + P_{st} R_{st}) \quad \ldots (4)$$

where:

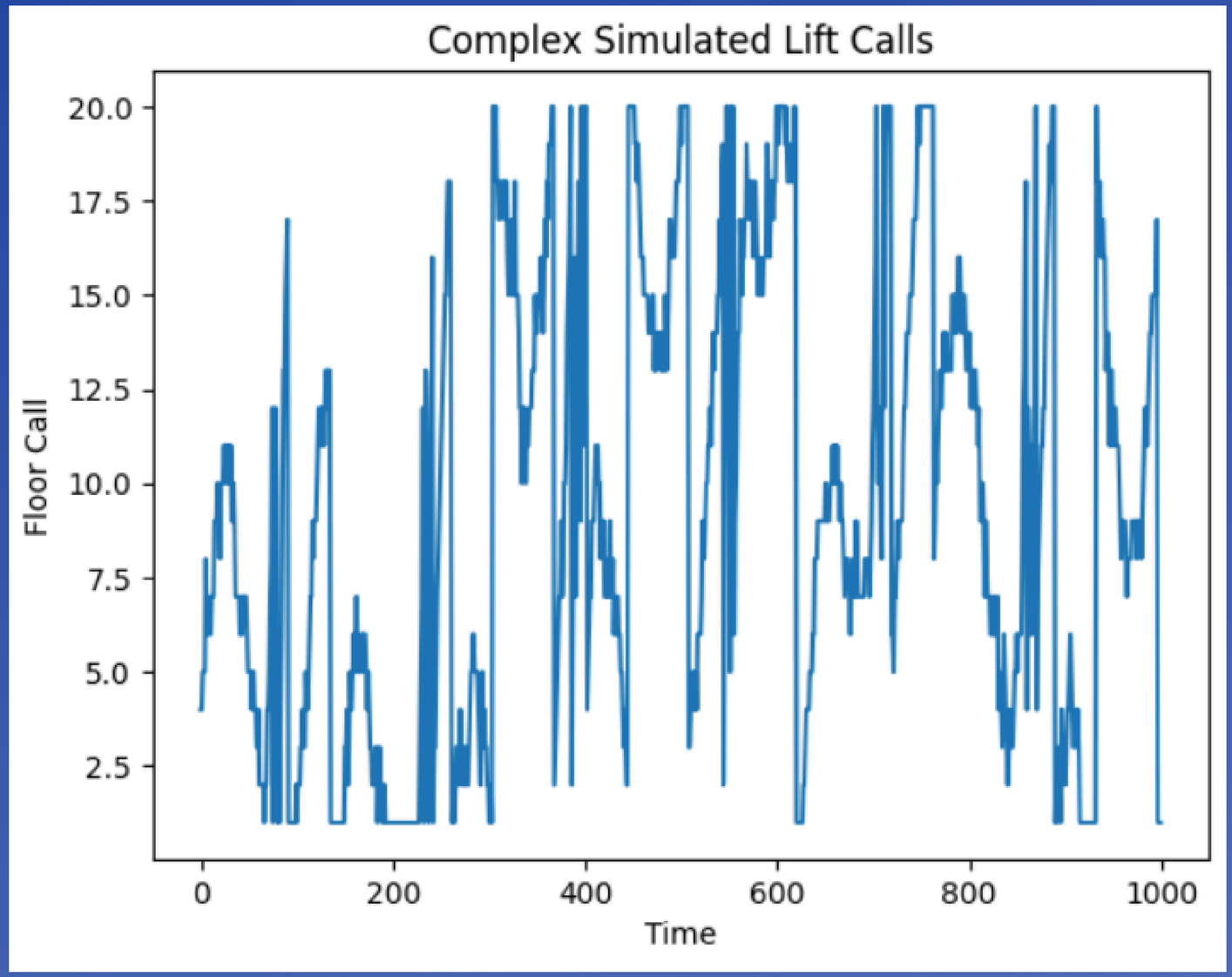
P_{id} is the power used when the lift is in idle mode (W)
(measured after door operations have ceased when stopped at a landing.)
P_{st} is the power used when the lift is in standby mode (W)
(measured after 5 minutes of inactivity.)
R_{id} is the ratio of idle time when consuming P_{id} (value<1)
R_{st} is the ratio of standby time when consuming P_{st} (value<1)
t_{av} is the time to travel the average travel distance for the target installation, including door times (s)

IMPLEMENTATION SO FAR...

- Devised a method to synthesize lift calls (simulation pre-requisite)
- Simulated two lift models-
  - Current lift system (based on FCFS)
  - Smart lift using Recurrent Neural Network
    - LSTM (Long Short-term Memory) model with a dense layer
    - ReLU activation function
    - Training on 1000 data points (lift calls)
    - Prediction on previous 20 points in the series
- Compared the results to understand future direction of the research work

SYNTHESIZED 1000 LIFT CALLS (SINGLE LIFT, 20 FLOORS) - USED FOR INITIAL ANALYSIS

- Random floor calls (0-20) were generated over time (0-1000).
- Floor selection was randomized using uniform/normal distributions.
- Time intervals between calls were varied to simulate natural behavior.
- Periodic spikes in lift usage were added to mimic peak hours.
- Data was smoothed using moving averages to resemble real-world scenarios.



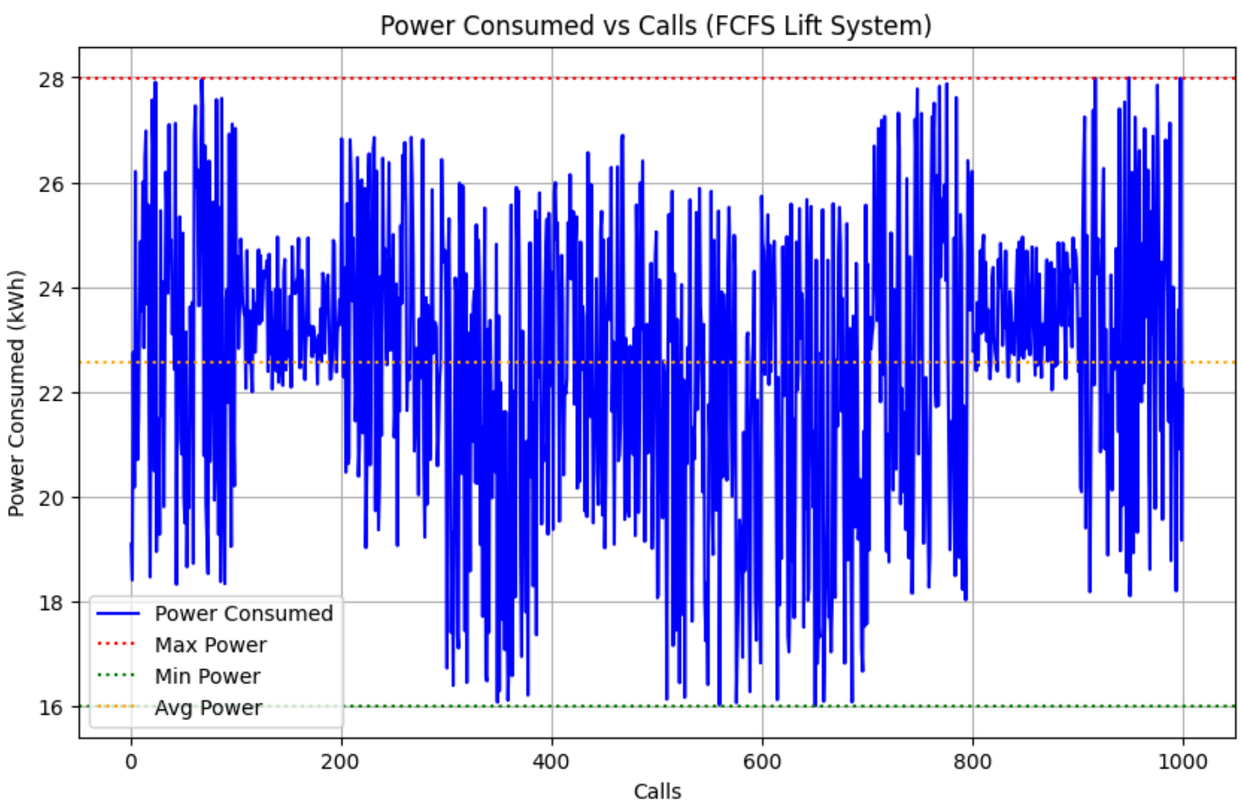
FCFS BASED LIFT SYSTEM MODEL

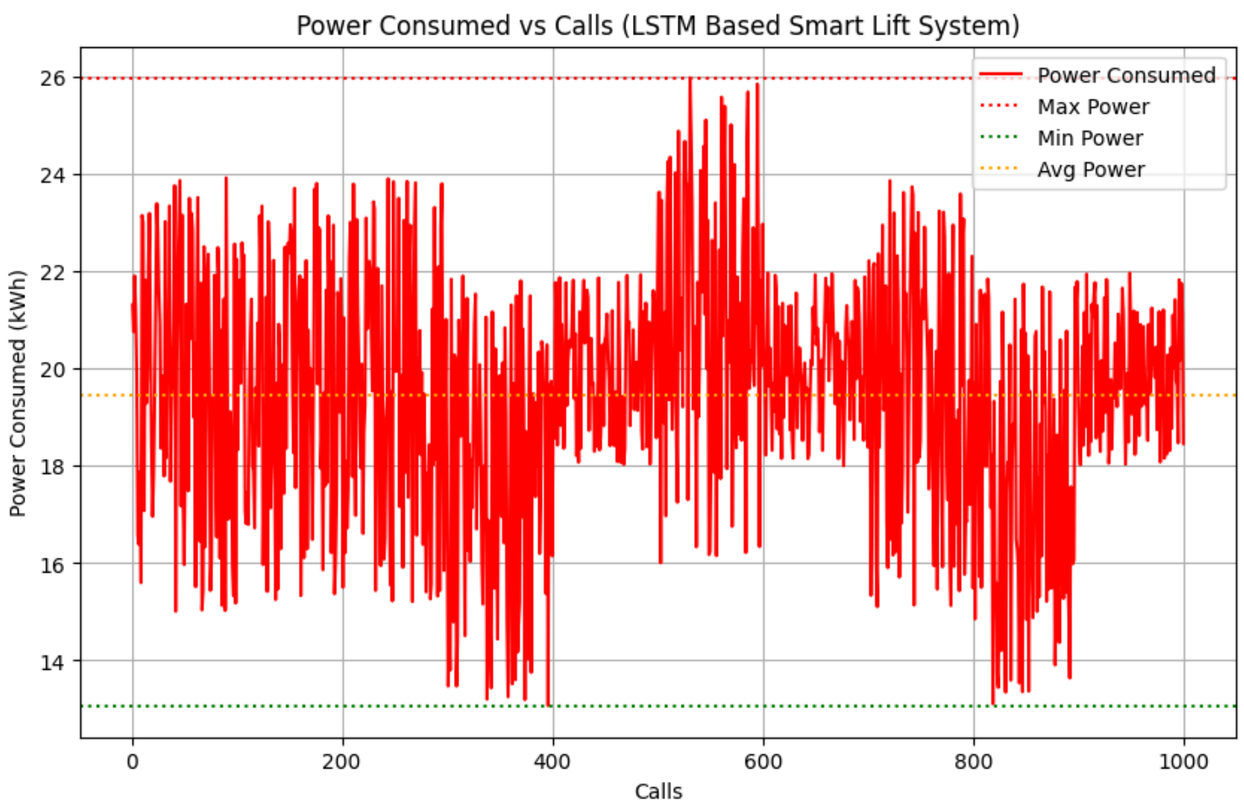
- The lift responds to floor calls sequentially, based on the order of requests.
- Calls in the path are considered if lift is not at capacity
- Calls were stored in a queue, processed one by one until completion.
- The lift moves directly to the next requested floor without optimization.
- Time taken for each floor transition was calculated and logged.
- Energy consumption was calculated for each floor movement based on distance.
- Idle energy usage was tracked when the lift was stationary between calls.
- Total energy usage was calculated after each complete trip or idle period.
- Python was used for simulation, handling queuing, movement, and energy tracking.

LSTM BASED SMART LIFT SYSTEM MODEL

- The LSTM model was built with 2 hidden layers, each containing 64 units.
- The input was a sequence of previous floor calls, with a 20-step window used to predict the next call.
- Adam optimizer and mean squared error (MSE) loss function were used for training.
- The model achieved an MSE of 4.215 after training on simulated lift call data.
- Predictions were utilized to optimize lift movements, reducing unnecessary stops and improving efficiency.
- Energy consumption was calculated for each floor transition based on movement and idle time.
- The LSTM-based system showed a significant reduction in energy usage compared to the FCFS model.

SIMULATION RESULTS - COMPARING POWER CONSUMED BY EACH MODEL

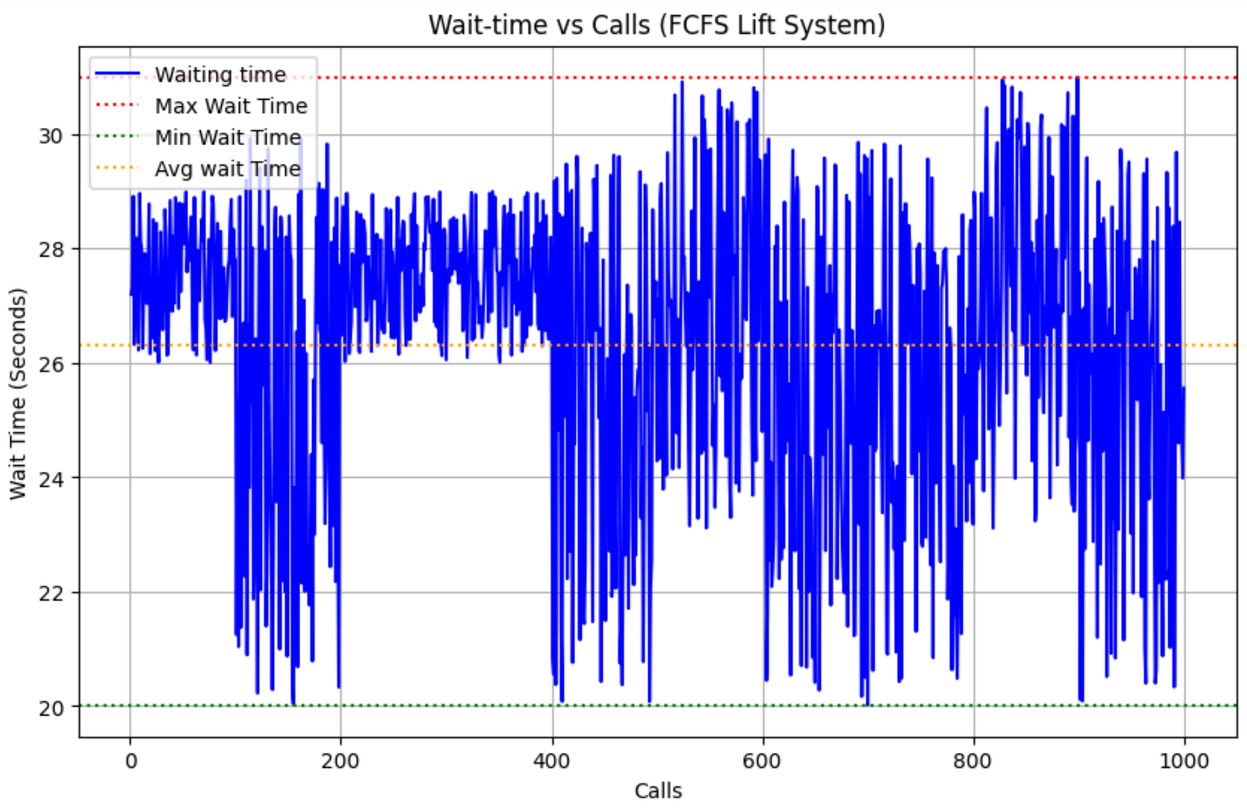


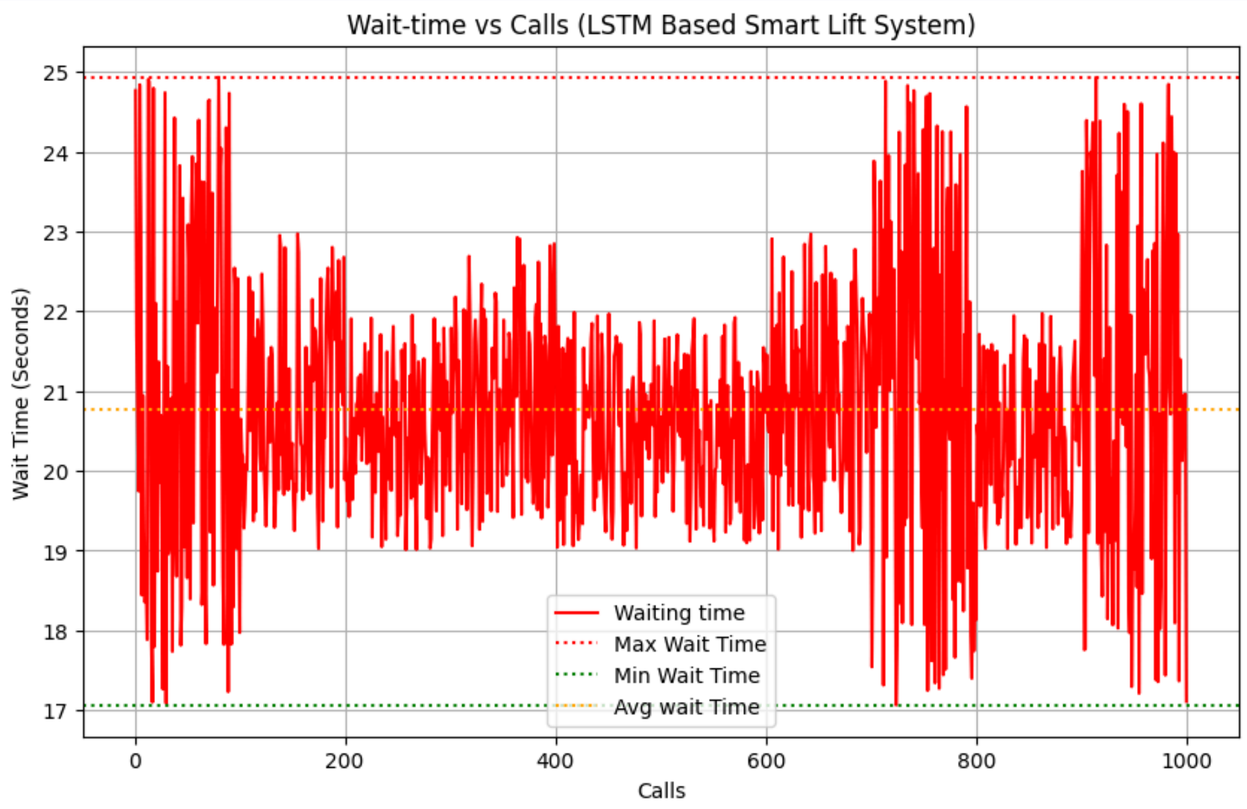


AVERAGE: 22.57 KWH
MINIMUM: 16 KWH
MAXIMUM: 28 KWH

AVERAGE: 19.46 KWH
MINIMUM: 13 KWH
MAXIMUM: 26 KWH

SIMULATION RESULTS - COMPARING WAIT TIME IN EACH MODEL





AVERAGE: 26 SECONDS
MINIMUM: 20 SECONDS
MAXIMUM: 31 SECONDS

AVERAGE: 20 SECONDS
MINIMUM: 17 SECONDS
MAXIMUM: 25 SECONDS

TO CONCLUDE...

- Proposed LSTM based Smart Lift System does show improvements, though not exceptional
- While the range of variation in terms of power and time both seem similar in the considered models, averages differ significantly, showing global improvement
- With further training, there is a scope to devise models that may prove efficient in varied settings

FURTHER WORK

- To build a dedicated lift simulation software that would be able to perform analysis of the models on the go, thus, boosting the research work
- To work with more complex lift calls' data
- To approach the problem of energy optimization for multi-lift systems

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THANK YOU